

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) FMM — Semester I Examination, 2022-23

Paper No. BCF-116: Business Statistics

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer all questions. All questions carry equal marks (14 marks each).

Question 1. “Statistics is not a science, it is a scientific method.” Discuss critically this statement and explain the utility and limitations of Statistics. [14]

OR

Rearrange the following series with equal intervals and then prepare 'less than' and 'more than' cumulative frequency distribution: [14]

oprule Class Interval	Frequency
0-5	3
5-6	2
6-9	7
9-12	5
12-17	16
17-18	12
18-20	15
20-24	20
24-25	8
25-30	10
30-36	2
Total	100
ottomrule	

Question 2. Define Geometric Mean and discuss its merits and demerits. Give two practical situations where you will recommend its use. [14]

OR

Find out mode and median wages from the following data: [14]

oprule Wages (extcurrency)	0-10	10-20	20-30	30-40	40-50
No. of Workers	5	7	15	25	8
ottomrule					

Question 3. Define Standard Deviation. Why is standard deviation regarded as superior to other measures of dispersion? Discuss. [14]

OR

Goals scored by teams A and B in a football season were as follows: [14]

No. of Goals Scored in a Match	No. of Matches	
	Team A	Team B
0	27	17
1	9	9
2	8	6
3	5	5
4	4	3

By calculating the coefficient of variation in each, state which team may be considered more consistent.

Question 4. What do you mean by index number? What points should be taken into consideration in the construction of index numbers? Explain. [14]

OR

From the following average prices of the commodities (given in rupees per unit), find chain base index no. with 2018 as base year: [14]

Items	2018	2019	2020	2021	2022
I	8	10	12	15	18
II	4	5	8	10	12

Question 5. What is time series analysis? Point out its importance. [14]

OR

Fit a straight line trend by the method of least squares from the following data: [14]

Year	2010	2011	2012	2013	2014	2015	2016
Production ('000 Ql.)	80	90	92	83	94	99	92

- Fit a straight line by the least squares method and calculate the trend values.
- Also estimate the production for 2018.
- Show the above series and the trend on graph paper.